

RE- Plate

Reinventing what goes on the plate



INDEX

1. Introduction.....	
2. Objective.....	
3. Data Sources	
4. Data Output & Key Deliverables.....	
5. Data Lakehouse Architecture.....	
6. Data Pipeline Design.....	
7. Pipeline Orchestration.....	
8. Pipeline Failure Handling.....	
9. Conclusion.....	

INTRODUCTION

Food waste is a serious issue worldwide, impacting both the planet and the economy. While large amounts of food are discarded daily, many people still lack access to fresh and healthy produce. RePlate offers a smart, cloud-based solution aimed at solving this issue by building a more sustainable and efficient food distribution system.

By using cloud computing, artificial intelligence, and real-time data, RePlate brings together farmers, sellers, and customers on one platform. It helps reduce food waste, lower costs, and make sure quality food reaches those who need it. With a flexible design and strong data tools, RePlate improves how food is managed, shared, and delivered—helping create a smarter and more sustainable food network.

Objective

RePlate aims to create a smarter and more sustainable food system by reducing waste and improving how food is shared. The platform focuses on connecting farmers, vendors, and customers through a cloud-based solution that supports real-time data and intelligent forecasting. By using automation and modern technology, it makes food management more efficient and cost-effective. At the same time, RePlate ensures that all information is kept secure and meets important food safety standards.



RE-Plate System – At a Glance

The RE-Plate system connects all parts of the food supply chain to reduce waste and improve food distribution.

- **Farmers & Restaurants:** Provide food data.
- **Website/App:** Access platform easily.
- **Payments:** Manage transactions.
- **Warehouse & Inventory:** Track stock and delivery.
- **Waste Metrics:** Measure food saved.
- **System Monitoring:** Ensure smooth operation.
- **Customer Feedback:** Improve service.
- **Reports:** Help marketing and growth.
- **Shipping:** Deliver food where it's needed.

Data Sources

These are the main places where RE-Plate collects data from:

- **Farmers & Restaurants:** Information about available surplus food.
- **Website & App Portal:** User activity, requests, and donations.
- **Payment Systems:** Details of financial transactions and records.
- **Warehouse Stock:** Real-time data on food availability and storage.
- **Customer Feedback:** Suggestions and satisfaction ratings from users.
- **System Monitoring Tools:** Performance and error tracking.

Key Deliverables

These are the important outcomes RE-Plate provides:

- **Waste Reduction Metrics:** Reports showing how much food waste is saved.
- **Inventory & Logistics Tracking:** Real-time updates on where food is stored and how it moves.
- **Shipping Coordination:** Ensures food gets delivered to the right places on time.
- **Marketing & Growth Reports:** Insights to help improve and scale the platform.
- **Improved Access to Food:** Making sure extra food reaches people in need efficiently.

Data Output:

RE-Plate uses advanced tools and technologies to turn raw data into useful, real-time insights. Each department benefits from specific data outputs that support smarter decisions and better services:

Inventory & Logistics

- **How It's Delivered:** Interactive dashboards (Power BI), geospatial data via Cosmos DB, and predictive models using machine learning (ML).
- **What It Delivers:** Real-time inventory visibility, efficient delivery route planning, and demand forecasting to reduce waste and improve food flow.

Shipping

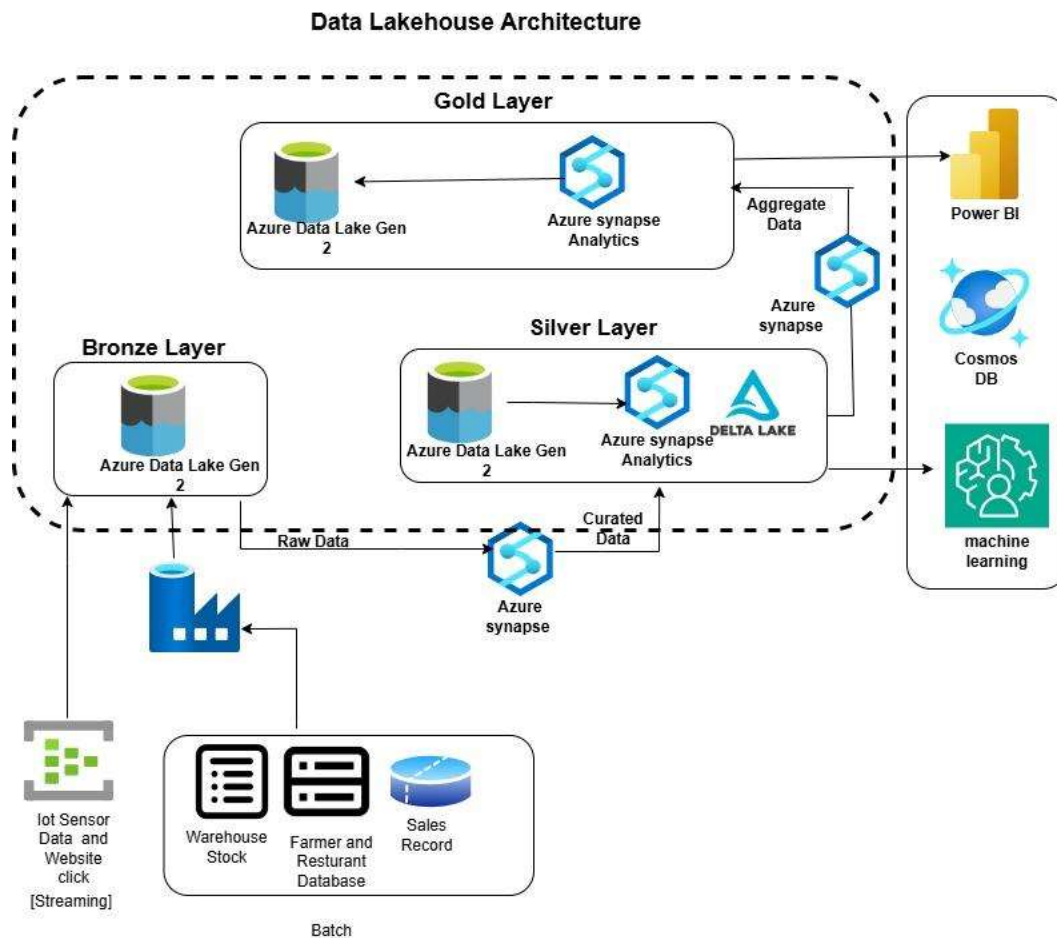
- **How It's Delivered:** Dispatch analytics dashboards, order logs, and Estimated Time of Delivery (ETD) algorithms powered by ML.
- **What It Delivers:** Ensures deliveries are on time, tracks performance of delivery partners, and helps meet service-level agreements (SLAs).

Marketing & Growth

- **How It's Delivered:** ROI-focused dashboards, customer profiles stored in Cosmos DB, and churn prediction models.
- **What It Delivers:** Helps track how well campaigns perform, understand different user types, and predict customer loyalty or drop-off rates.

Customer Feedback

- **How It's Delivered:** Sentiment analysis dashboards, review storage, and natural language processing (NLP) for automated analysis.
- **What It Delivers:** Highlights customer emotions and trends, auto-detects common issues, and provides useful insights to improve user experience.



HOW IT WORKS:

1. Bronze Layer – raw Data Collection

- **Sources:** Data comes from sensors (like IoT), website clicks, warehouse systems, sales records, and farmer/restaurant databases.
- **Type:** It collects both *real-time (streaming)* and *batch* data.
- **Tool:** This raw data is stored in Azure Data Lake Gen2.

2. Silver Layer – cleaned & organized Data

- **Purpose:** Here, raw data is cleaned, organized, and processed (called **curated data**).
- **Tools Used:**
 - **Azure Synapse Analytics:** For processing and analytics.

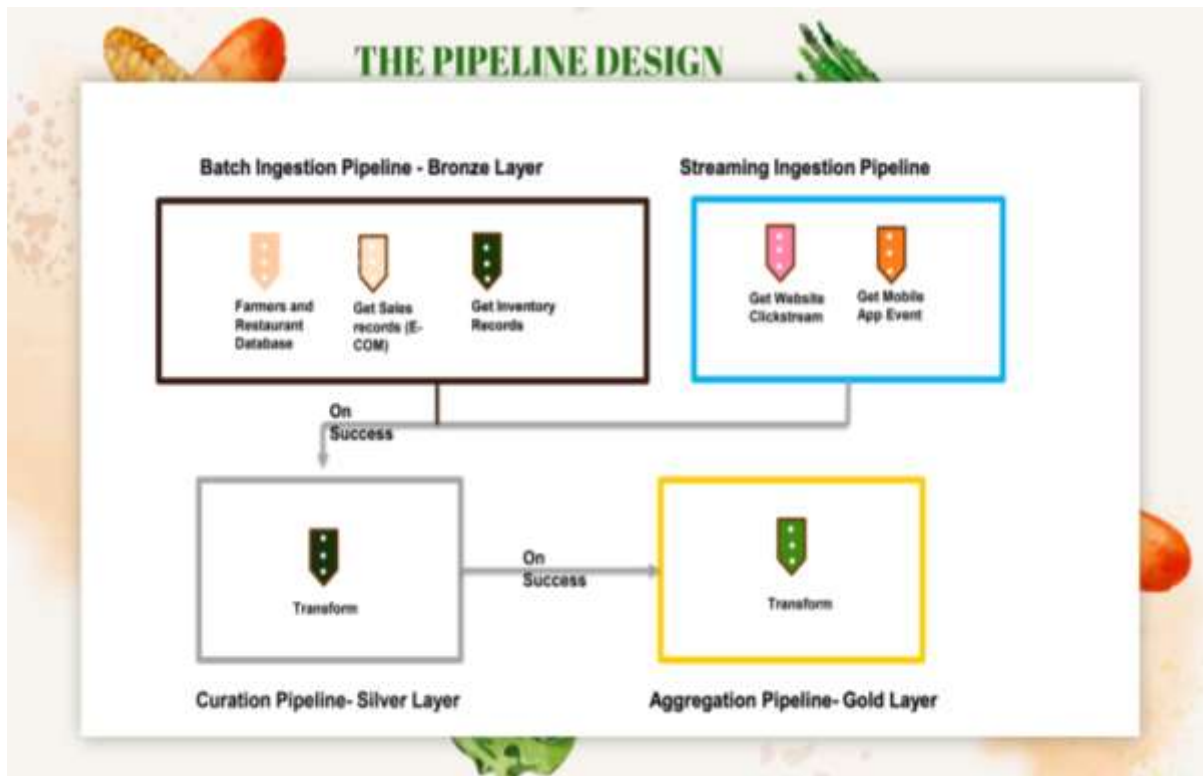
- **Delta Lake:** For storing curated data with support for updates and deletes.

3. Gold Layer – Ready for Analysis

- **Purpose:** Final version of the data, well-organized and aggregated (summarized).
- **Use:** This layer is used for dashboards, reports, predictions, etc.
- **Tools Used:**
 - **Power BI:** To create visual dashboards and reports.
 - **Cosmos DB:** For storing data related to customer profiles or app behavior.
 - **Machine Learning Models:** To find patterns and make predictions.

Key Tools Involved

- **Azure Data Lake Gen2:** Stores huge amounts of data.
- **Azure Synapse:** Analyzes and manages data.
- **Power BI:** Displays results in easy-to-understand visuals.
- **Cosmos DB:** Stores data for fast access in apps.
- **Machine Learning:** Helps with intelligent insights and predictions.



Overview of the Pipeline Design

This system is divided into three main stages, also called layers:

1. Bronze Layer – Data Ingestion

This is where raw data is first collected from different sources using two types of pipelines:

Batch Ingestion Pipeline

- **Farmers and Restaurant Database** – Info about food sources.
- **Sales Records (E-COM)** – Data from online sales.
- **Inventory Records** – Current stock levels.

These are collected in batches (at set times).

Streaming Ingestion Pipeline

- **Website Clickstream** – Tracks what users click on websites.
- **Mobile App Events** – Tracks actions on mobile apps.

These are collected in real-time (streamed instantly).

2. Silver Layer – Curation Pipeline

- After raw data is collected, it is sent to the transform stage.
- This is where data is cleaned, structured, and prepared for deeper analysis.
- Only successful data from the bronze layer moves here.

3. Gold Layer – Aggregation Pipeline

- The curated data is transformed again into a more summarized and refined format.
- This version is ready for business insights, dashboards, and machine learning.
- It's the final and most valuable form of the data.

Pipeline Orchestration:

Master Pipeline:

Think of this as the main controller. It's like the manager that tells other smaller pipelines (sub-pipelines) when and how to do their job.

Scheduling:

- Uses Azure Data Factory to run tasks at set times.
- Uses Event Hub to react to real-time events.

- This ensures that data flows automatically and on time.

Dependency Chaining:

- The process happens step-by-step:
 1. First, the Bronze layer collects raw data.
 2. Then, the Silver layer cleans and organizes it.
 3. Finally, the Gold layer prepares it for reporting and dashboards.
- Each step waits for the previous one to finish to avoid errors.

Purpose:

- To make sure data is collected, cleaned, and prepared in a smooth, automated way.
- It helps deliver high-quality data that's ready to be used for business decisions.

Sub-Pipelines (Layers):

- **Bronze Layer:** Collects raw data from apps, websites, and databases.
- **Silver Layer:** Cleans and organizes the data so it's easy to use.
- **Gold Layer:** Turns data into useful charts, dashboards, and business insights.

Pipeline Failure Handling:

1.Failure Causes:

- Data source connectivity issues
- Corrupted or incomplete data files
- Hardware or network interruptions

2.Configuration Errors:

- Incorrect pipeline settings or missing parameters
- Misconfigured credentials or endpoints
- Invalid data mapping rules

3.Retry Strategy:

- Automatically retry on failure after a short wait
- Use exponential backoff to gradually increase wait time
- Log each retry attempt for monitoring and analysis

4.What If All Retries Fail?

- Send alert notifications to the operations team
- Automatically roll back to the last successful checkpoint
- Trigger manual intervention for further investigation

CONCLUSION:

RePlate is a smart system that uses cloud technology to help reduce food waste and make the food supply process better. It collects data from many sources, cleans it, and turns it into useful information for making decisions.

By using tools like Azure Data Factory, Event Hub, and Power BI, RePlate can track food, manage inventory, and give real-time updates. This helps farmers, vendors, and customers work together more easily.

The system is built to grow in the future, with plans to use IoT devices like drones and better AI models. RePlate shows how technology can solve real problems and create a more sustainable way to share food.